UNO

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	1024 megabytes

Troubadour_Ggmz and Fengmi123 are playing a special UNO game. UNO cards have four colors: red, yellow, green, and blue, denoted by `R`, `Y`, `G`, and `B`, respectively. Each color contains n cards, with ranks being any integers from 1 to n; there may be two cards with the same color and rank. The whole deck has $4n$ cards.

At the beginning of the game, Troubadour_Ggmz arranges all $4n$ cards in a row, then Fengmi123 can perform the following operation any number of times: choose two adjacent cards, and if they are the same color or the same rank, swap their positions.

The goal of the game is to rearrange all cards into a state that is both color-ordered and rank-ordered, i.e.:

- The color order is `R` $<$ `Y` $<$ `G` $<$ `B`, meaning all red cards appear before all yellow cards, all yellow cards appear before all green cards, and all green cards appear before all blue cards;
- For cards of the same color, the sequence of ranks must be non-decreasing.

Fengmi123 wants to know whether she can achieve the target state through a sequence of allowed operations (possibly zero).

Input

There are multiple test cases. The first line contains an integer t ($1 \le t \le 1000$), the number of test cases. For each test case:

The first line contains a positive integer n ($1 \le n \le 1000$).

Then $4n$ lines follow, each containing a positive integer x_i and a character c_i, where $1 \le x_i \le n$, $c_i \in \{\texttt{R}, \texttt{Y}, \texttt{G}, \texttt{B}\}$, indicating the rank and color of the i-th card in order. It is guaranteed that each color appears exactly n times, but cards with the same color and rank may appear multiple times.

It is guaranteed that the sum of n over all test cases does not exceed 1000.

Output

For each test case, if the target state can be reached, output a line containing "`YES`", otherwise output a line containing "`NO`".

You may output the answer in any case (upper or lower). For example, the strings "`yEs`", "`yes`", "`Yes`", and "`YES`" will all be recognized as positive responses.

Example

standard input	standard output
2 2 1 R 2 Y 2 R 1 G 1 Y 2 B 2 G 1 B 2 1 R 2 Y 1 Y 2 B 2 R 1 G 2 G 1 B	YES NO